



5464 - Uncorrelated Noise and Optimal Extraction of JWST/NIRSpec Sources using PypeIt

Cycle: 3, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Debora Pelliccia (PI)	University of California - Santa Cruz
Dr. Kyle Westfall (CoI)	University of California - Santa Cruz
Prof. Joseph Hennawi (CoI)	University of California - Santa Barbara
Dr. Jason X. Prochaska (CoI)	University of California - Santa Cruz
Dr. John M. O'Meara (CoI)	California Association for Research in Astronomy (CARA)
James Davies (CoI) (ESA Member)	Max Planck Institute for Astronomy

OBSERVATIONS

ABSTRACT

Combining spectral images by first resampling dithered observations into a regular grid (which is the standard JWST calwebb_spec3 reduction procedure) introduces correlation between pixels that smooths the data and complicates their noise properties. Particularly for observations that push JWST's sensitivity limits, this approach can affect the detection of key spectral features, such as the Lyman break in spectra of galaxies at extreme redshift, or even lead to false detections. Many studies have used custom codes and applied empirical corrections to deal with the correlated noise, affecting the reproducibility of the results.

In this archival research proposal, we request support to add JWST NIRSpec to the pool of spectrographs supported by the spectroscopic data reduction pipeline PypeIt. PypeIt is an open-source Python package that performs end-to-end data reduction for more than 50 ground-based spectrographs, and provides routines that yield coadded spectral images without correlated pixels by instead using an irregular wavelength grid. PypeIt is largely used in the astronomy community (with > 500 users and ~150 publications, so far); therefore, enabling it to reduce JWST data will provide a familiar output datamodel for current PypeIt users, and facilitate the synergy between JWST and ground-based observations. The JWST-

specific script will be distributed and documented with the main PypeIt repository and, like for all the supported spectrographs, the following products will be provided: 2D spectral images with uncorrelated noise, robust 1D boxcar and optimal extractions, and detailed QA plots that allow users to quickly assess the results.

OBSERVING DESCRIPTION